Analysis and Prediction of Monsoon Hazards in Nepal Using Machine Learning Approaches

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***Abstract*—Floods, landslides, heavy rainfall and lightning kill several hundred people in Nepal every monsoon. Nepal's new anticipatory action framework releases relief before disasters strike once triggers (pre-agreed forecast conditions) are met, so it needs forecasts of when harm is likely as well as where hazards can occur. The problem is a daily, district-level forecast: for each of the 77 districts and each day from April to October, whether a damaging incident (a death, missing person, injury or destroyed house) will be recorded in the next three days. The dataset, built from the official disaster record (BIPAD, snapshot of 19 September 2026) and ERA5 reanalysis rainfall, holds 146,223 district-days for 2017 to 2025, 6.8% of them positive. K-means, Ward and Gaussian mixture clustering group districts into risk regions. Logistic regression, random forest, gradient boosting and support vector machines are tuned with season-blocked cross-validation, calibrated on 2023 and tested once on 2024 to 2025. District-by-month climatology scores 0.143 in precision-recall AUC (PR-AUC); the best models reach 0.177 (95% block-bootstrap interval 0.141 to 0.216), a small gain that is significant for three of five models on test, while in a 2026 check all five beat climatology, three significantly. The six regions are moderately stable and interpretable but add nothing to a pooled model that already encodes district-by-month history. A perfect three-day rain forecast raises PR-AUC to 0.21 to 0.22, more than any change of model, so better rain forecasts matter more than model choice. The ranking can help prioritise readiness under a district alert budget; reporting bias and lightning are the main caveats.**

***Index Terms*—impact-based forecasting, machine learning, clustering, landslides, floods, lightning, Nepal**

I. Introduction

Nepal's monsoon brings most of the country's annual rain between June and September and, with it, most of its disaster deaths. In late September 2026, heavy rain set off floods and landslides that killed at least 21 people [1]. On 11 October 2026 a flood from a suspected landslide blockage near the Budhi Gandaki in Gorkha struck on a dry day, sweeping away houses and several bridges, with at least five people missing [2], [3]. Extreme rainfall has increased at many stations, with spatially mixed trends [4]–[6], and the Himalayan arc holds a large share of the world's fatal landslides [7], [8]. Recorded harm also depends on exposure, construction and reporting [9]. Nepal's National Anticipatory Action Framework (2026) aims to release money and relief before a hazard strikes (anticipatory action), with forecasts and triggers as one of its three building blocks [10]. A trigger is a pre-agreed forecast condition that releases this support. For a district administration deciding whether to pre-position relief, the useful trigger is a forecast of impact: whether people are likely to be hurt in the next few days, whatever the mechanism.

Machine learning has been applied widely to Nepal's hazards, mostly in static susceptibility mapping, such as national landslide maps from inventories and terrain [11] and multi-hazard assessments under climate scenarios [12], [13]. Such maps show where hazards can occur but not when. Rainfall thresholds supply timing, but for slope failure alone [14], [15]. Impact-based forecasting with machine learning exists for single hazards in data-rich settings, such as global landslide nowcasting [16], severe-weather and flood impacts in Catalonia [17], [18], and tropical-cyclone housing damage [19]. The impact module of BIPAD, Nepal's national disaster portal, was prototyped for riverine floods at two stations [20]. To the author's knowledge, no published model forecasts, district by district and day by day, whether any of the main monsoon hazards will cause casualties or destroy homes in Nepal. A second gap is methodological. Nepal's districts range from the Terai plain to the high Himalaya, so a single national model may hide where it fails. Yet comparative evidence from hydrology suggests that pooled models, trained on all sites together, often outperform regional ones [21], [22]. Whether learned regions help a harm forecast has not been tested.

This study builds and evaluates a daily, district-level harm forecast for Nepal's monsoon hazards from open data. The objectives are: (1) to group Nepal's districts into monsoon risk regions, groups of districts with similar impact history and rain climate, using clustering; (2) to predict by classification whether a district will suffer a damaging hydro-meteorological incident in the next three days; and (3) to test whether adding the risk regions improves prediction, and to explain which factors drive risk. A damaging incident is one that records at least one death, missing person, injury or destroyed house.

The first contribution is a harm dataset built from BIPAD whose unit is the district-day (one district on one day). Changes in recording are made explicit, and it is checked for leakage (use of information unavailable at forecast time). To the author's knowledge it is the first multi-hazard dataset of its kind for Nepal. The second is a fair test of whether learned regions help, against the climatology (historical district-month

rate), persistence and threshold baselines a forecaster would use; the regions add almost nothing. The third is an analysis of where and why the forecast fails, by region, hazard and month. Two bounds, no weather and a perfect rain forecast, separate the value of rainfall information from that of the model.

II. Related Work

A hazard forecast says whether a physical event is likely; an impact forecast says what it will do to people and assets [23]. Impact-based warnings aim at the second [24], and anticipatory action uses them to release funds before an event [25]. Monsoon harm in Nepal is multi-hazard in the sense of Gill and Malamud [26]. Floods, landslides and heavy rain share a driver and often coincide, while lightning shares the season but not the mechanism, a framing increasingly adopted in risk research [27]–[29]. A target built from damaging incidents is a harm label. It folds hazard, exposure and vulnerability together as recorded, which suits an anticipatory trigger but makes the label sensitive to reporting.

A. Rainfall Thresholds and Early Warning

Intensity-duration thresholds for landslide initiation are the oldest operational tool [15], [30], [31]. For Nepal, Dahal and Hasegawa fitted $I = 73.90\,D^{-0.79}$ (mm/h over D hours) to 193 landslides, which gives about 144 mm/day [14], [32]. Thresholds are transparent, but they say nothing about whether anyone is hurt. They also transfer poorly from rain gauges to the coarse grid of a reanalysis, a model reconstruction of past weather. This study uses the Nepal threshold as a baseline and, for fairness, refits a percentile threshold on ERA5 rain.

B. Machine Learning Hazard Mapping in Nepal

Reichenbach et al. reviewed statistical susceptibility mapping and its reliance on inventories [33]. Kincey et al. mapped rainfall-triggered landslide susceptibility and exposure for Nepal, showing that susceptibility peaks in the High Himalaya but exposure in the Middle Hills [11], [34]. Thapa et al. cluster Nepal into zones with K-means on static susceptibility layers [12], and Shrestha et al. assess compound flood-landslide and drought-fire interactions under CMIP6 scenarios [13]. These studies locate hazard; none forecasts daily harm. Since Thapa et al. already cluster Nepal, clustering in this study is a means to test whether regions help a forecast.

C. Forecasting Occurrence and Impact with Machine Learning

Stanley et al. turned NASA's landslide hazard assessment into a data-driven global nowcast [16]. Their XGBoost model uses antecedent rain and static factors and is trained on inventories with a temporal hold-out. Doerksen et al. predict landslides by district in Nepal over 7, 10 and 14-day windows, using random forests and U-Nets on satellite precipitation and geomorphic data [35]. Dahal et al. model landslide occurrence and density in space and time for the Gorkha-affected region [36], the closest Nepal comparator.

Impact forecasting from impact records is most developed in Korea, where heavy-rain damage is classified from daily weather [37]–[39], and in Catalonia, where Iglesias et al. predict severe-weather impact on people at municipality-day level and beat the official warning [17]. Morales et al. predict flood-related emergency calls per municipality and hour, with walk-forward evaluation and separate models per population-density group [18]. Operational anticipatory-action triggers are event-based. Kooshki Forooshani et al. built a global tropical-cyclone housing-damage model that is used to trigger anticipatory action [19], and Sedhain et al. show that such triggers can have very low detection on a real typhoon [40]. This study therefore reports detection and false alarms at a fixed alert budget (a cap on the share of district-days alerted). Table I compares the closest works. None combines several hazards, a daily horizon and a national loss record for Nepal.

TABLE I
Closest works and how this study differs. Every forecasting row whose design is known uses time-ordered evaluation; Thapa et al. is static. The Doerksen et al. and Dahal et al. rows are taken from their abstracts.

Study	Unit, horizon	Hazards	Label	Baseline
Stanley et al. 2021 [16], global	30 arc-second (about 1 km) grid, daily nowcast	Landslide	Landslide inventories	LHASA 1.0
Doerksen et al. 2023 [35], Nepal	District, 7, 10 and 14-day windows	Landslide	Landslide occurrence	RF and U-Net compared; thresholds critiqued
Dahal et al. 2024 [36], Gorkha region	1 km cell, about 6-month periods	Landslide	Occurrence and areal density	Not in abstract
Iglesias et al. 2024 [17], Catalonia	Municipality, daily	Severe weather	Emergency-incident counts	Official warning
Morales et al. 2026 [18], Catalonia	Municipality, next hour	Flood	112 emergency calls	Climatology-based warnings
Kooshki Forooshani et al. 2024 [19], Philippines	Municipality and 0.1° grid, per event	Tropical cyclone	Humanitarian damage data, 39 typhoons	Earlier operational model
Thapa et al. 2026 [12], Nepal	District and grid, static	Flood, landslide	Susceptibility layers	None
This study, Nepal	District, 3 days	Flood, landslide, heavy rain, lightning	BIPAD loss record	Climatology, persistence, thresholds

D. Disaster Loss Records and Their Biases

Loss databases under-record small and remote events, and their coverage changes over time, so trends in recorded losses mix changes in hazard, exposure and reporting [41]–[43]. In Nepal, recorded lightning deaths rose with internet access [44], and Harvey et al. document gaps in landslide reporting [45]. BIPAD is the Government of Nepal's incident portal, maintained by the National Disaster Risk Reduction and Management Authority (NDRRMA) [46], [47]. It is the only national, incident-level source with locations and losses, so these biases shape what any model can learn.

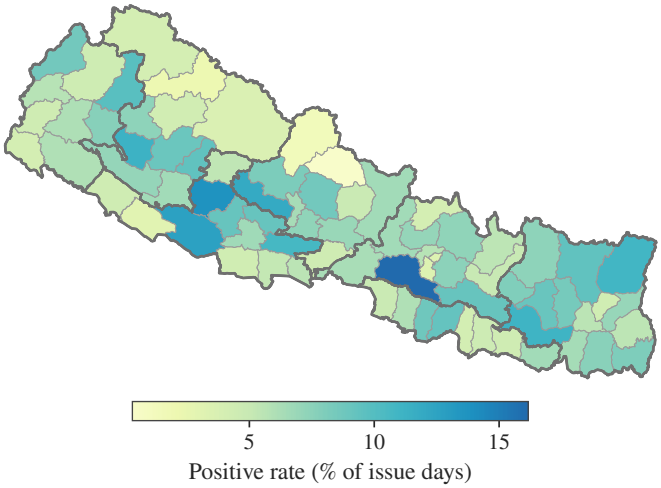


Fig. 1. Share of issue days with a damaging incident in the next three days, by district, 2017 to 2025. Boundaries from OCHA COD-AB [48]. The belt map is Fig. 10 in the appendix.

III. Problem and Dataset

A. Study Area and Season

Nepal's 77 districts span three ecological belts within 200 km: the Terai plain below 300 m, the Hill belt of the Mahabharat and Middle Hills, and the Mountain belt above 3,000 m. This study uses an elevation proxy for the belts. A district is Mountain if its mean ward elevation is at least 1,700 m and at least 5% of its wards lie above 3,000 m, Terai if the mean is below 700 m and fewer than 15% of wards lie above 1,000 m, and Hill otherwise. Rainfall is strongly orographic and varies across the country [49]. Issue days, at the end of which a forecast is made, run from 1 April to 28 October so that the three-day target window stays inside April to October. The pre-monsoon months are included because the lightning season runs from March to August [50] and lightning harm in BIPAD peaks in April and May, while floods, landslides and heavy-rain damage peak in July and August (Fig. 7, appendix).

B. Target Data: the BIPAD Incident Record

The label comes from a snapshot of the BIPAD incident API taken on 19 September 2026, holding 63,549 incidents of all types [46]. Of these, 16,790 records dated 2011 to 2026 have the hazard title Flood, Landslide, Heavy Rainfall or Thunderbolt and are kept. BIPAD itself links each incident to a ward. The ward, municipality and district tables were downloaded once, with their dates, so each incident's district cannot change when the portal updates. Every incident maps to a district. Timestamps are converted to Nepal time and truncated to the calendar day. The loss fields are used only to set the damaging flag and are dropped before any feature is built.

The record has two entry periods. Incidents from 2016 to 2019 were entered in bulk years later, with a median entry lag of 930 days for 2017 incidents. From 2020 the median lag is one day and 94% of incidents are entered within three days. Recent-incident features may therefore use only what was entered by the issue day. Reporting also grew: between 2017 and 2024 the share of district-days with any recorded incident rose 2.8 times, but the share with a damaging incident rose only 1.4 times (Fig. 8, appendix). Because the growth concentrates in minor incidents, the target is restricted to damaging ones.

TABLE II
Season-blocked splits and positive rates.

Split	Seasons	Rows	Positives	y (%)	y_{rain} (%)
Train	2017–2022	97,482	6,482	6.65	4.26
Validate	2023	16,247	722	4.44	2.57
Test	2024–2025	32,494	2,749	8.46	5.49
Check	2026 (to 16 Aug)	10,626	614	5.78	3.39

C. Predictors

Daily weather is ERA5 reanalysis [51] from the Open-Meteo historical API [52] at each district centroid, for 2 March to 31 October of 2016 to 2026 (201,971 district-days, no gaps). The variables are precipitation sum, precipitation hours and maximum temperature in Nepal time. ERA5 has a 0.25° grid, so each district is one point and orographic peaks are smoothed. Terrain is summarised from the elevations of all 6,803 ward centroids (Copernicus DEM through Open-Meteo) as district mean elevation, elevation range and the share of wards above 1,000 m and 3,000 m.

D. Problem Definition and Sampling

The unit is a district d on an issue day t. With $D_{d,\tau} = 1$ if district d has at least one damaging incident of the four hazards dated day τ, the primary, all-hazard target is

$$y_{d,t} = \max\left(D_{d,t+1},\, D_{d,t+2},\, D_{d,t+3}\right). \tag{1}$$

A three-day window matches a short anticipatory-action lead time and absorbs the day-level uncertainty of incident records. Secondary labels take the same form for each hazard, for fatal incidents, and for the rain-driven hazards alone (flood, landslide and heavy rainfall, y_{rain}). The panel of all district-days holds 77 districts and 211 issue days per season for 2017 to 2025: 146,223 rows, 9,953 of them positive ($y = 1$, 6.81%). Lightning is part of 38% of positive rows and the only hazard in 36% of them. The 2026 season is an out-of-time check; its issue days stop at 16 August, so the last window ends 31 days before the snapshot. Table II gives the season-blocked splits, which assign whole seasons to training, validation or test. The validation year has the lowest positive rate and the test years the highest, which matters for calibration.

E. Data Quality

Within a district, consecutive issue days share two of their three target days, so y has a lag-one autocorrelation of about 0.69, and one storm can hit many districts on the same day (36 on 28 September 2024). Tuning folds and bootstrap blocks are therefore whole seasons and whole weeks, never single rows.

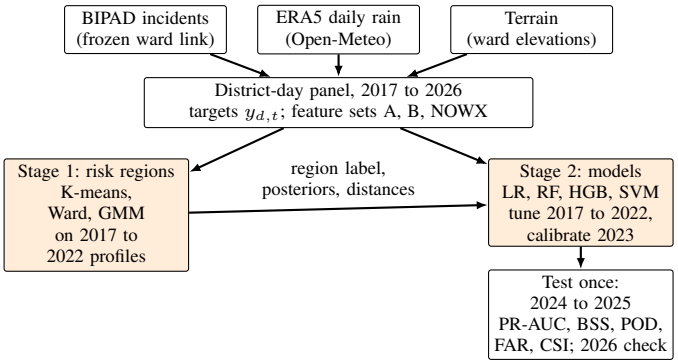


Fig. 2. Two-stage cluster-then-classify pipeline with the time-ordered evaluation.

IV. Methods and Experimental Setup

A. Design

Fig. 2 shows the two-stage design. Stage 1 clusters the districts into risk regions from their training-season profiles, and Stage 2 trains models on the district-day panel with and without the regions as features. The regions enter one pooled model as extra features, and no per-region models are fitted. Since set A (below) already holds district-by-month history, the regions can help only if they carry information that history lacks. In hydrology, models trained on many catchments learn regional behaviour on their own [21], [22], and Piernik and Morzy found that a bare cluster label is the weakest way to pass cluster information to a model and that tree ensembles rarely benefit [53], so a null result for objective 3 would be informative. The expectation was registered before the test seasons were scored: if regions helped at all, they would help logistic regression and not the trees.

B. Stage 1: Regionalisation

Each district is described by 37 numbers from the training seasons 2017 to 2022 only. They are the monthly rate of damaging days for each hazard (4 hazards $\times$ 7 months), the monthly mean daily ERA5 rain (7) and two terrain values; no exposure or vulnerability layer is used. The profile uses same-day rates, so it describes history and does not contain the forecast target. The columns are standardised and reduced by principal component analysis (PCA). The first four components carry 49% of the variance and each later one adds less than 6%, so these four form the clustering space. The 19-component space that reaches 90% is a sensitivity check.

K-means [54] with 50 restarts, Ward agglomerative clustering and a diagonal Gaussian mixture model [55] are fitted for $k = 2$ to 8. Validity is scored with the silhouette [56], Davies–Bouldin [57] and Calinski–Harabasz [58] indices and the mixture's BIC. The rule for k was fixed in advance: among K-means solutions whose smallest region has at least five districts, take the best silhouette, with Davies–Bouldin as tie-break. Stability is the adjusted Rand index (ARI) [59] between the reference labels and labels refitted on 200 bootstrap resamples of the six seasons. The same index measures agreement with the ecological belts and the seven provinces.

Following Piernik and Morzy [53], the regions are passed to Stage 2 in three forms: the K-means label (one-hot), the mixture's posterior probabilities and the distances to the K-means centroids.

C. Stage 2: Classification

Four classifier families are compared, all in scikit-learn [60]: logistic regression [61] on standardised inputs, random forest [62], histogram gradient boosting [63], and support vector machines (SVMs). The SVMs are a linear SVM on all rows and an RBF SVM on a stratified subsample of 12,000 rows. Class imbalance is handled with class weights, since resampling distorts calibrated risk [64]. Hyperparameters (Table XII, appendix) are tuned on leave-one-season-out folds within 2017 to 2022, following the season-blocked logic of Roberts et al. [65]. The tuning score is mean PR-AUC, the area under the precision-recall curve (Section IV-E). The chosen setting is refitted on 2017 to 2022, and isotonic regression fitted on 2023 maps its score to a probability. Because 2023 has the lowest positive rate of any season, a second, pooled calibrator fitted on the out-of-fold scores of 2017 to 2023 is reported as a post hoc sensitivity. The "best model" for the detailed analyses is the one with the highest tuning PR-AUC, chosen before the test seasons were scored.

Three feature sets are used (Table XI, appendix). Set A holds what is known at the end of day t: ERA5 rain over 1 to 30 days, an antecedent index and the district percentile of 3-day rain, rain hours, maximum temperature, season, terrain, damaging incidents known at t in the district and its neighbours, and shrunken district-by-month history rates per hazard. ERA5 is released days after the fact, so set A is a hindcast: it replays what a perfect same-day weather analysis would have known, which no operational forecast has. Set B adds the observed ERA5 rain for $t+1$ to $t+3$, a perfect-rain-forecast reference at ERA5 resolution used only as a ceiling. NOWX (no weather) drops every weather column and is the floor. NOWX to A measures the value of rainfall information, and A to B the value of a perfect three-day rain forecast.

D. Baselines

Five baselines are scored on the same test rows:

- climatology, the district-by-month positive rate of 2017 to 2022, which is also the reference for skill scores [66];
- persistence, an alert if a damaging incident dated $t-3$ to t was already entered by day t;
- the Dahal and Hasegawa thresholds applied to ERA5, 144 mm on day t or 182 mm over 72 hours, both from the same curve [14];
- a refitted ERA5 threshold, the 3-day rain percentile cut with the best critical success index on 2017 to 2022;
- a year-only control, a logistic regression on year and month that shows whether year alone ranks district-days; no model uses year as a feature.

E. Metrics and Uncertainty

The primary metric is PR-AUC, computed as average precision, which is appropriate at a 7% base rate [67]. Probabilistic skill is measured by the Brier score [68] and the Brier skill score (BSS) against climatology. Operational value is read at a fixed share of 5% of the evaluated test district-days, the top-ranked ones, which averages about four district alerts a day over the season. Tied scores at the cut are broken at random, and each rule baseline is compared with models re-thresholded to the rule's own alert rate. The probability of detection (POD) is the share of positives alerted and the false alarm ratio (FAR) the share of alerts that are false. The critical success index (CSI) is hits divided by hits, misses and false alarms [69]. Ranking metrics use the raw score and Brier metrics the calibrated probability. All intervals are 95% percentile intervals from a block bootstrap over ISO weeks (1,000 resamples); they condition on the two test seasons and do not capture variation between seasons. Differences between models, feature sets or baselines use the same resamples, which gives paired intervals. Feature effects are examined with SHAP [70] and permutation importance.

F. Tools and Leakage Controls

Everything runs in Python 3.12 with scikit-learn 1.9.1 in five Jupyter notebooks with a fixed seed. Leakage is treated as the main threat to validity [71]. Splits are by season only, and loss fields never enter features. Set A is checked to be unchanged when future weather is scrambled and not-yet-entered incidents are removed. Fitted quantities use training seasons only, with leave-one-season-out history encoding for training rows. Recent-incident counts use only incidents entered by day t, with seeded synthetic entry lags before 2020. The appendix lists the automated checks.

V. Results

A. Exploratory Findings

For 2017 to 2025, BIPAD holds 5,203 damaging incidents of the four hazards: 1,788 lightning, 1,606 landslide, 1,046 heavy rainfall and 763 flood. The positive rate rises monotonically with the 3-day rain percentile, from 3.3% in the lowest decile to 14.7% in the highest. The curve is almost identical in the test seasons (Fig. 9, appendix). Hill districts have the highest positive rate (7.7%); Makwanpur and Rolpa (Hill) and Dang (inner Terai) exceed 12% (Fig. 1).

B. Risk Regions

Table III gives the validity indices. Cluster structure is moderate at best; the silhouette never exceeds 0.33. The pre-registered rule selects $k = 6$ for K-means (silhouette 0.327, Davies–Bouldin 0.93, smallest region six districts). Ward and the mixture agree with it (ARI 0.75 and 0.73), although BIC alone would favour three components. Under season resampling K-means is moderately stable (bootstrap ARI 0.55, 5th to 95th percentile 0.37 to 0.73), Ward slightly less so (0.47) and the mixture poorly (0.21). The regions do not reproduce the belts (ARI 0.24) or the provinces (0.22). Fig. 3 maps them and Fig. 11 (appendix) profiles them:

- region 2, the central and western Terai (12 districts), with July and August flood and heavy-rain damage;
- region 0, the eastern and inner Terai, with the highest lightning rates;
- region 3, the dry far west and north-west (20 districts);
- region 4, the central Hill belt including the Kathmandu valley;
- region 1, six wet eastern and central Hill and Mountain districts with a July landslide peak;
- region 5, seven landslide-prone Hill districts with the highest training damage rate (9.9%).

On the test seasons the regions still separate positive rates (7.0% to 10.7%), so the ablation below is a fair test.

TABLE III
Cluster validity for $k = 2$ to 8 in the four-component PCA space, and stability at the chosen $k = 6$.

Method	k	Silhouette	DB	CH	Min. size
K-means	2	0.279	1.355	32.1	37
	3	0.276	1.210	31.9	23
	4	0.318	1.022	33.6	9
	5	0.310	1.067	34.0	8
	6	0.327	0.932	34.6	6
	7	0.328	0.886	35.1	3
	8	0.330	0.876	34.1	2
Ward	2	0.272	1.361	30.8	34
	3	0.264	1.177	30.8	20
	4	0.308	1.011	32.5	9
	5	0.303	1.082	33.3	8
	6	0.324	0.956	32.6	3
	7	0.327	0.889	33.0	3
	8	0.314	0.920	32.7	3
GMM	2	0.138	2.436	10.2	37
	3	0.178	1.652	15.6	12
	4	0.263	1.297	22.0	6
	5	0.291	1.049	26.6	3
	6	0.305	0.872	29.4	3
	7	0.302	0.869	29.8	1
	8	0.316	0.891	30.9	2

At $k = 6$: bootstrap ARI [5th, 95th]; ARI vs belts / provinces			
K-means	6	0.55 [0.37, 0.73]	0.24 / 0.22
Ward	6	0.47 [0.30, 0.65]	0.23 / 0.24
GMM	6	0.21 [0.05, 0.46]	0.25 / 0.22

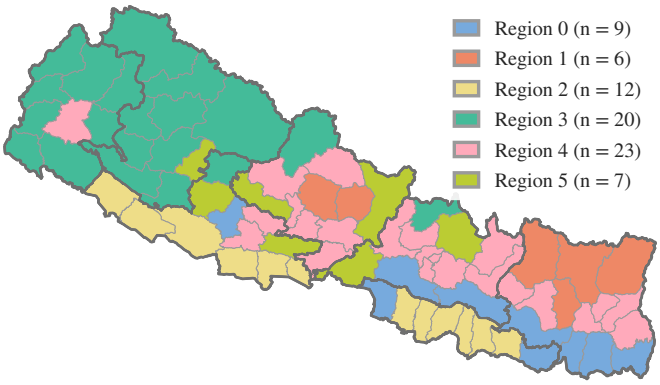


Fig. 3. The six K-means risk regions fitted on 2017 to 2022 profiles.

C. Baseline Comparison and Main Result

Tables IV and V give the main result on 2024 to 2025. District-by-month climatology already ranks well (PR-AUC 0.143, 95% interval 0.120 to 0.166) because harm clusters in a few districts and months. Persistence is worse (0.101): only one in five damaging incidents is known on the day it happens, and most harm does not repeat within three days. The Dahal and Hasegawa thresholds fire on 1.4% of ERA5 district-days and catch 4% of positives, so gauge thresholds do not transfer to a 0.25° grid. The refitted percentile threshold does no better than climatology (0.150). The year-only control (0.128) has no district term, so it falls below climatology by construction. Adding year to a district-history control changes PR-AUC by only +0.002 (+0.000 to +0.004; appendix), so year carries almost no ranking signal.

All four classifier families (five models, two of them SVMs) beat climatology on PR-AUC with set A. Logistic regression and the linear SVM reach 0.177 and gradient boosting 0.175. For logistic regression the paired gain is +0.035 (+0.010 to +0.066), and the gain is significant for three of the five models. The tuning rule had selected the random forest as best (mean fold PR-AUC 0.150 against 0.136 for logistic regression). On test it scores 0.165, and the interval of its gain just includes zero. Paired gaps between the four main families are at most 0.012 (logistic regression over the random forest, -0.001 to $+0.026$), so logistic regression is best only post hoc, and within the grids searched the families are close; the subsampled RBF SVM is weaker. No interval is corrected for multiple comparisons, so the pattern across families matters more than any single model's gain. Most of the gain comes from 2024 (Table VIII). ROC-AUC rises from 0.65 for climatology to 0.69 to 0.71. The models separate from climatology mainly at recall below 0.5 (Fig. 4(a)), the operating range of an alert system.

Probabilistic skill depends on the calibrator. With isotonic calibration on 2023 alone, BSS is near zero or slightly negative for every model. Mean predicted probabilities are 0.047 to 0.052 against an observed 0.085, because 2023 had half the test years' positive rate (Fig. 4(b)). The pooled calibrator lifts BSS to 0.010 to 0.020 and the mean prediction to 0.066 to 0.068. At the 5% budget (Fig. 4(c)), logistic regression detects 14.2% of positive district-days with a FAR of 0.76, against 11.5% and 0.80 for climatology, and CSI rises from 0.078 to 0.098. At each rule's own alert rate (Table VI), logistic regression and the random forest detect more than all three rules, for persistence 20.6% and 19.0% against 17.2%, though only slightly more at the rare Dahal and Hasegawa rate. A threshold fixed on 2023 scores would instead have alerted 6.3% (random forest) and 8.6% (logistic regression) of test district-days, more in 2024 than in 2025.

D. Ablations: Weather, Perfect Forecast and Regions

Table XV (appendix) separates the sources of skill, with paired intervals. Without any weather column (NOWX), logistic regression, gradient boosting and the linear SVM still reach 0.164 from season, terrain, history and recent incidents.

TABLE IV

RANKING SKILL ON TEST 2024 TO 2025 (32,494 DISTRICT-DAYS, BASE RATE 0.085): PR-AUC AND PAIRED DIFFERENCE AGAINST CLIMATOLOGY (CLIM.), WITH 95% WEEKLY BLOCK-BOOTSTRAP INTERVALS. PERSIST.: PERSISTENCE; D&H THR., ERA5 THR.: DAHAL AND HASEGAWA AND REFITTED THRESHOLDS; YEAR CTRL.: YEAR-ONLY CONTROL; LR, RF, HGB, SVM-L, SVM-RBF: THE FIVE MODELS.

Model	PR-AUC [95% CI]	Δ vs Clim. [95% CI]
Baselines		
Clim.	0.143 [0.120, 0.166]	
Persist.	0.101 [0.082, 0.121]	-0.042 [-0.059, -0.025]
D&H thr.	0.091 [0.072, 0.112]	-0.051 [-0.072, -0.030]
ERA5 thr.	0.150 [0.118, 0.182]	+0.008 [-0.021, +0.036]
Year ctrl.	0.128 [0.100, 0.160]	-0.015 [-0.034, +0.007]
Models, set A (known at t)		
LR	0.177 [0.141, 0.216]	+0.035 [+0.010, +0.066]
RF	0.165 [0.130, 0.202]	+0.023 [-0.002, +0.052]
HGB	0.175 [0.139, 0.211]	+0.032 [+0.008, +0.060]
SVM-L	0.177 [0.141, 0.215]	+0.034 [+0.010, +0.065]
SVM-RBF	0.153 [0.125, 0.186]	+0.010 [-0.013, +0.039]
RF floor and ceiling		
RF, no weather	0.143 [0.117, 0.172]	+0.001 [-0.014, +0.016]
RF, set B	0.219 [0.156, 0.299]	+0.077 [+0.018, +0.153]

TABLE V

PROBABILISTIC AND OPERATIONAL SKILL ON TEST 2024 TO 2025. BSS WITH THE 2023 OR THE POOLED CALIBRATOR; POD, FAR AND CSI AT THE 5% BUDGET.

Model	BSS (2023)	BSS (pooled)	POD	FAR	CSI
Clim.	0.000		0.115	0.805	0.078
Year ctrl.	-0.006		0.103	0.826	0.069
LR	0.005	0.018	0.142	0.760	0.098
RF	-0.006	0.016	0.119	0.799	0.081
HGB	-0.006	0.020	0.131	0.778	0.090
SVM-L	-0.015	0.018	0.141	0.762	0.097
SVM-RBF	-0.010	0.010	0.120	0.796	0.082
RF, no weather	-0.017		0.104	0.825	0.070
RF, set B	0.028		0.172	0.710	0.121

The random forest without weather falls to climatology. Set A adds +0.011 to +0.022 to NOWX, with a paired interval that excludes zero only for the random forest. The perfect three-day rain forecast of set B adds another +0.034 to +0.054 for the four main families (0.211 to 0.220), with every interval above zero, and raises POD at the 5% budget to 0.17. This gain is about three to four times the largest paired gap between the main families.

The regions do not help (Table VII). For logistic regression every encoding is within ± 0.004 of the model without region features, and the one-hot label is slightly worse. For the random forest, the posterior and distance encodings add +0.004 to +0.005 with intervals that include zero. A plain district one-hot adds +0.005 (+0.002 to +0.008), as much as any region encoding. The pre-registered expectation that regions would help the linear model more than the trees is not supported. Set A already carries district-by-month history, and the regions add nothing beyond it.

E. Seasons, the Rain-Driven Target and the 2026 Check

Skill varies by season with the base rate (Table VIII). The models gain most in 2024, the test year with most harm

TABLE VI
RULE BASELINES ON TEST 2024 TO 2025 AT THEIR OWN ALERT RATE, AGAINST CLIMATOLOGY, LOGISTIC REGRESSION AND THE RANDOM FOREST (SET A) RE-THRESHOLDED TO THE SAME RATE.

Rule (alert rate)		Rule	Clim.	LR	RF
Persist. (8.1%)	POD	0.172	0.181	0.206	0.190
	FAR	0.820	0.811	0.785	0.801
	CSI	0.096	0.102	0.118	0.108
D&H thr. (1.4%)	POD	0.041	0.031	0.047	0.044
	FAR	0.755	0.812	0.716	0.737
	CSI	0.036	0.028	0.042	0.039
ERA5 thr. (21.3%)	POD	0.391	0.397	0.452	0.434
	FAR	0.845	0.843	0.821	0.828
	CSI	0.125	0.127	0.147	0.141

TABLE VII
OBJECTIVE 3: PR-AUC ON TEST WITH THE FOUR LOCATION ENCODINGS ADDED TO SET A, AND PAIRED DIFFERENCES AGAINST THE SAME MODEL WITHOUT THEM (RF: RANDOM FOREST; LR: LOGISTIC REGRESSION).

Model	Encoding	PR-AUC	Δ vs none [95% CI]
RF	none (set A)	0.165	
	+ K-means label	0.166	+0.001 [-0.002, +0.004]
	+ GMM posteriors	0.170	+0.005 [-0.001, +0.011]
	+ centroid distances	0.169	+0.004 [-0.002, +0.011]
	+ district one-hot	0.170	+0.005 [+0.002, +0.008]
LR	none (set A)	0.177	
	+ K-means label	0.174	-0.004 [-0.008, +0.001]
	+ GMM posteriors	0.176	-0.002 [-0.006, +0.002]
	+ centroid distances	0.178	+0.000 [-0.005, +0.005]
	+ district one-hot	0.175	-0.003 [-0.009, +0.003]

(base rate 0.090, with the late-September floods), where the random forest reaches 0.202 and logistic regression 0.214, against 0.158 for climatology. In 2025 the random forest is below climatology and logistic regression level with it. On the 2026 check (issue days to 16 August, base rate 0.058), every model is above climatology (0.100 to 0.108 against 0.087) and has positive BSS (0.011 to 0.027). The paired PR-AUC interval excludes zero only for the random forest, gradient boosting and the RBF SVM; for the two best models on test it includes zero. On the rain-driven target, which leaves out lightning, climatology falls to 0.123 at a base rate of 0.055. Logistic regression reaches 0.177 with set A (paired gain +0.055, +0.019 to +0.096) and 0.229 with set B, a larger relative gain from rainfall information.

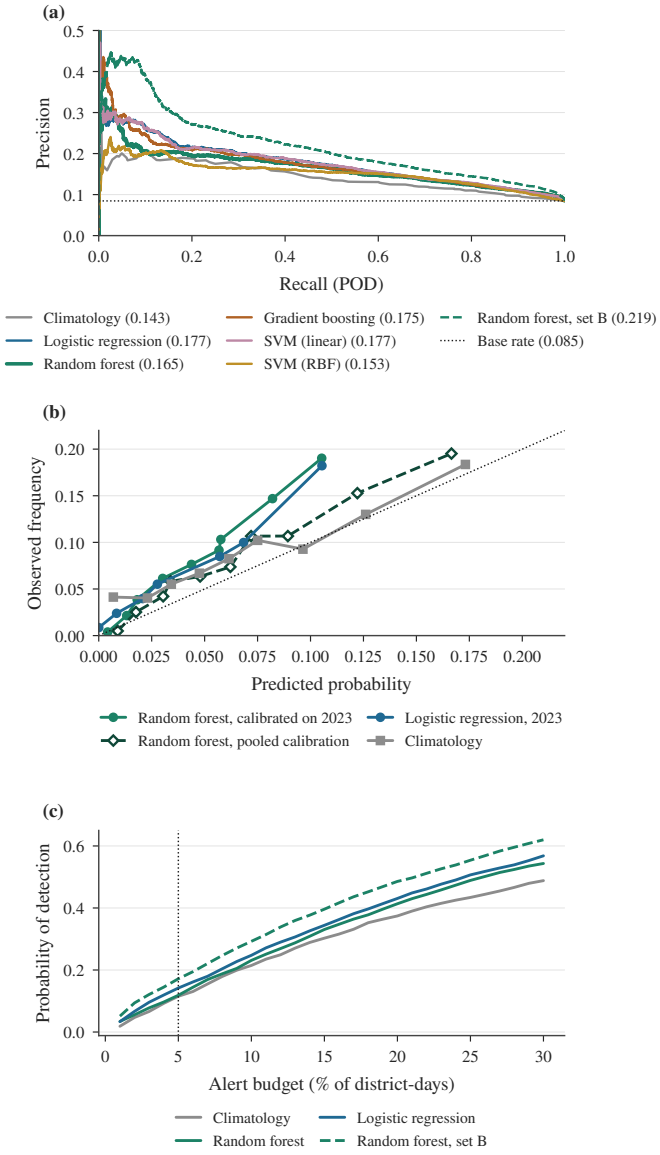


Fig. 4. Test 2024 to 2025. (a) Precision-recall curves of the set A models, climatology and the random forest with set B. (b) Reliability of the random forest and logistic regression probabilities; calibration on 2023 alone under-predicts. (c) Probability of detection against the alert budget; the dotted line is the 5% budget.

F. Where the Forecast Fails and What Drives It

The forecast is most useful where rain-driven harm dominates and least where lightning does. Table IX and Fig. 12 (appendix) break the random forest down by region, belt, hazard and month. By region, the gain over climatology is largest in the central and western Terai (region 2: 0.206 against 0.166) and the central Hill belt (region 4: 0.139 against 0.109). The six wet eastern districts of region 1 score below climatology (0.151 against 0.173) and receive few alerts (POD 0.04). A national budget also spreads alerts unevenly: it alerts 9% of rows in regions 0 and 5 but about 3% in region 3 and 2% in the Mountain belt, which argues for regional budgets. By hazard, the all-hazard model ranks floods, heavy rainfall and landslides better than climatology and lightning worse (0.038 against 0.043). The perfect-forecast ceiling roughly doubles the first three but leaves lightning almost unchanged (0.045), consistent with its convective mechanism. By month, detection peaks in July (POD 0.22) and is weakest in April and May (0.03 to 0.06), when lightning dominates and the model issues almost no alerts. In August the model under-performs climatology (0.164 against 0.191), as positives spread over many districts after a wet July.

Fig. 5 shows what the random forest uses; no single feature dominates. By mean absolute SHAP value the 3-day rain percentile (0.025), rain on day t (0.023) and the district history rate (0.020) lead. High rain percentiles, high history rates and

TABLE VIII

PR-AUC PER SEASON FOR CLIMATOLOGY, THE RANDOM FOREST AND LOGISTIC REGRESSION (SET A), WITH THE SEASON'S BASE RATE AND THE RANDOM FOREST'S BSS. THE LOWER ROWS REPEAT THIS FOR THE RAIN-DRIVEN TARGET.

Season	Role	Base	Clim.	RF	LR	RF BSS
2023	calib.	0.044	0.081	0.086	0.089	+0.034
2024	test	0.090	0.158	0.202	0.214	-0.008
2025	test	0.079	0.131	0.125	0.133	-0.004
2026	check	0.058	0.087	0.100	0.102	+0.027
Rain-driven target, test 2024 to 2025						
2024–25	test	0.055	0.123	0.163	0.177	+0.005
2024–25	set B			0.231	0.229	+0.044

TABLE IX

RANDOM FOREST (SET A) ON TEST 2024 TO 2025 BY REGION, BELT AND HAZARD: POSITIVES, BASE RATE, CLIMATOLOGY AND MODEL PR-AUC, AND POD AT THE NATIONAL 5% BUDGET. HAZARD ROWS SCORE THE ALL-HAZARD MODEL AGAINST EACH HAZARD'S OWN LABEL.

Stratum	Pos.	Base	Clim.	RF	POD
Region 0	350	0.092	0.145	0.162	0.160
Region 1	271	0.107	0.173	0.151	0.037
Region 2	353	0.070	0.166	0.206	0.218
Region 3	756	0.090	0.183	0.196	0.093
Region 4	735	0.076	0.109	0.139	0.095
Region 5	284	0.096	0.155	0.154	0.151
Terai	590	0.067	0.123	0.167	0.185
Hill	1500	0.096	0.146	0.163	0.119
Mountain	659	0.082	0.162	0.175	0.058
Flood (own label)	467	0.014	0.024	0.041	0.161
Landslide (own label)	827	0.025	0.051	0.062	0.143
Heavy rainfall (own label)	725	0.022	0.047	0.067	0.167
Lightning (own label)	1054	0.032	0.043	0.038	0.052

mid-season days push the score up. High elevation and high 30-day totals push it down, because the wettest late-season weeks fall after the damage peak. Permutation importance agrees on the rain features for the random forest but, for logistic regression, ranks season far higher (day-of-year sine 0.035). The linear model relies on a smooth seasonal curve, while the forest uses history and rain interactions.

G. Case Study: 26 to 28 September 2024

From 26 to 28 September 2024 a late-monsoon system stalled over central and eastern Nepal and brought record three-day rain to central Nepal [72]. Over those days BIPAD recorded 232 damaging incidents in 42 of the 77 districts (148 landslides, 47 floods, 35 heavy-rainfall and 2 lightning incidents), the worst event in the record. Fig. 6 shows the forecast issued at the end of 25 September. With set A, the random forest ranks the districts no better than climatology (PR-AUC 0.579 against 0.595). Six of its top 10 and 9 of its top 20 districts were hit, against 4 and 11 for climatology. Its probabilities stay below 0.25 everywhere, because ERA5 rain up to 25 September had not yet shown the extreme totals of 27 and 28 September. The set B version, which sees that rain, raises the whole central and eastern belt (PR-AUC 0.771; 8 of the top 10 and 15 of the top 20 hit) and leaves the dry west low. Most of the gain came from knowing the coming rain.

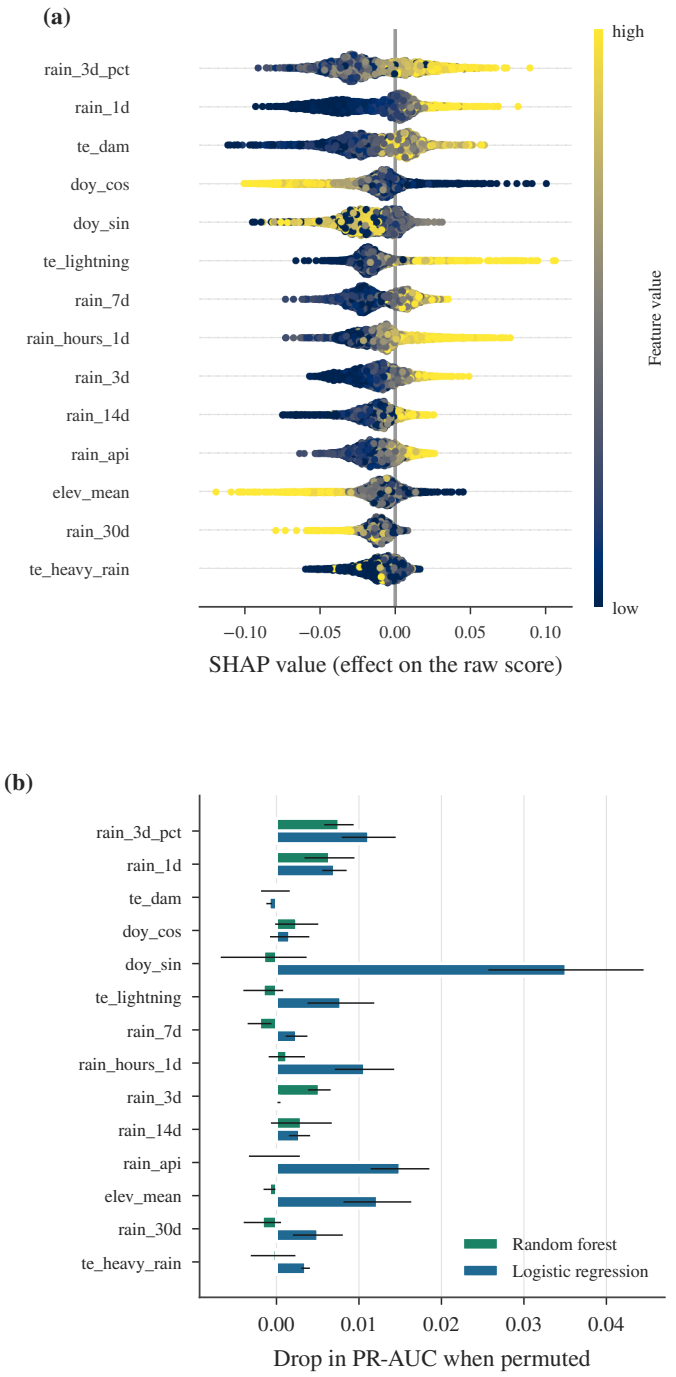


Fig. 5. (a) SHAP summary for the random forest (set A) on a stratified sample of 4,000 test rows. (b) Permutation importance (drop in PR-AUC) for the random forest and logistic regression on the same rows.

VI. DISCUSSION AND CONCLUSIONS

A. Summary and Comparison with the Literature

Objective 1 produced six moderately stable, interpretable monsoon risk regions that are not simply the belts or the provinces. Objective 2 produced a district-day harm forecast with real but small ranking skill, and objective 3 found that the regions add nothing to the models. Skill numbers are not

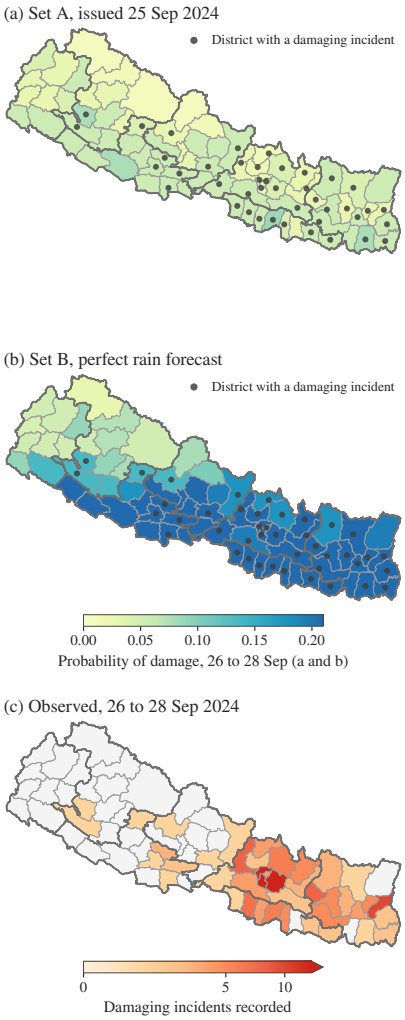


Fig. 6. Forecast issued at the end of 25 September 2024. (a) Random forest probability with set A; (b) with set B (perfect forecast). Dots mark districts that recorded damaging incidents on 26 to 28 September; (a) and (b) share one colour scale. (c) Damaging incidents per district on those days (scale capped at the 95th percentile).

directly comparable across studies because targets and base rates differ, but the pattern is. Stanley et al. about doubled the hit rate of a heuristic landslide nowcast [16], Iglesias et al. beat the official warning [17], and Morales et al. found that the choice of model matters little [18]. In all of them, as in this study, most of the predictable signal is spatial and seasonal, and the daily increment from weather is the hard part. The null result for regions is close to expected by construction, because the pooled model already encodes district-by-month history. It agrees with Piernik and Morzy's finding that cluster labels help little once informative features exist [53], but it does not test the pooled-against-regional comparison of hydrology [21], [22].

B. Interpretation

First, the floor is high and the ceiling low. Day-t reanalysis rain adds only 0.01 to 0.02 to the no-weather floor of 0.164, while a perfect three-day rain forecast adds 0.03 to 0.05 more, so the coming rain matters more than the rain already observed. This argues for archived operational forecasts and higher-resolution precipitation before further model tuning. Second, lightning weakens the all-hazard target. It is the only hazard in 36% of positives, peaks before the monsoon and is weakly tied to daily rain, so the model learns two hazard types at once. Third, calibration transfers poorly across seasons with different base rates: a calibrator fitted on the quietest season under-predicts the wettest. The pooled calibrator fixes most of this, and an operational system would recalibrate each season or use several years' climatology as its reference [66].

C. Trends in the Field and Their Limits Here

Hazard forecasting is moving from rainfall thresholds to learned models [16], [17], [35], including large-sample models that pool many sites [21], [22]. The results support better rain input, since a perfect three-day forecast adds more than any change of model, although reanalysis rain over the Himalaya is itself uncertain [73]. They give no reason for more model capacity: the four families are close, and a harm label that reflects reporting as much as weather [42], [45] caps what any model can extract, so a deeper network would mainly fit the record's biases more closely. For a record like BIPAD, label quality, exposure data and time-ordered evaluation of triggers [24], [71] are likely to return more than model complexity.

D. Practical Implications

Set A is a hindcast: ERA5 arrives about five days late, so none of its rain is available at issue time, and a FAR of 0.76 is too high for a cash-release trigger. The output suits a daily readiness and prioritisation ranking of districts under a fixed alert budget, for District Disaster Management Committees and local governments alongside the heavy-rain and flood warnings of the Department of Hydrology and Meteorology. At about four alerts a day it detects about 14% of damaging district-days with three false alarms per hit, against about 12% and four for climatology. The gain is modest and would be cheap once operational forecasts replace ERA5, since the other inputs are open. The per-region error table shows where local information should override the ranking.

E. Social, Ethical, Legal and Professional Considerations

A missed alert and a false alarm have different costs: the first can cost lives, the second credibility and, under anticipatory action, money released for nothing. The 5% budget used in this study favours detection, and a deployed system should set its budget with the agencies that act on it. The model learns from what BIPAD records. Report-based loss records tend to miss remote areas [41], [45], and Nepal's recorded lightning deaths rose with internet access [44]; this was not tested for BIPAD. A district with no record is not necessarily safe. Districts with weaker reporting would receive lower scores, so an alert system built on them could steer attention away from those who need it most. BIPAD is public, with no licence stated; ERA5 data are CC BY 4.0 through Open-Meteo, the

Copernicus DEM, also accessed through Open-Meteo, has its own licence, and COD-AB is CC BY-IGO. Loss counts are discarded after labelling, so no person is identifiable and no incident-level loss value is published. This is a research prototype, not an operational warning system, and NDRRMA remains the custodian of the record.

F. Limitations and Future Work

The label is a reporting record with two entry periods and growing coverage; the damaging filter and the year controls reduce but do not remove the effect. Set A uses the ERA5 analysis of day t, so its skill is an upper bound on what the same features would give operationally, and set B is explicitly a ceiling. One reanalysis point per district smooths orographic rain, and the free archive has no convective variables, which limits lightning. The project proposal planned slope from a 30 m DEM and the landslide susceptibility layer of Kincey et al. [11], [34]. Neither was used; terrain came from ward-centroid elevation, so slope enters only indirectly. Floods from suspected landslide blockages, such as the Gorkha flood, or from glacier failures can strike on dry days and are outside the model's scope. Future work should use archived operational forecasts at one to three day leads and add lightning observations or convective indices. It should also move to municipality units where the ward link allows, and add exposure so that the target can be weighted by people at risk.

G. Conclusion

This study built a leakage-checked, district-day harm forecasting dataset for Nepal's four main monsoon hazards from open data. It compared three clusterers and four classifier families with time-ordered evaluation, and measured each against baselines a forecaster would use. The models beat climatology on ranking by a small margin. The gain is significant for three of the five models on test, not including the pre-specified random forest, and positive for all five in the 2026 check, three significantly. Knowing the coming rain would help far more than changing the model. The regions, though moderately stable and interpretable, carry no information that district-by-month history lacks. The code, frozen data and run evidence are provided so that every number can be reproduced.

APPENDIX A
RUN EVIDENCE

A. Environment and Run Times

All runs were on an Apple M4 laptop (24 GB) under macOS 26.3 with Python 3.12.12, pandas 3.0.6, numpy 2.5.3, scikit-learn 1.9.1, matplotlib 3.11.2, seaborn 0.13.2, geopandas 1.2.0, shap 0.53.0 and pyarrow 26.0.0, pinned exactly in `uv.lock` (and in `pyproject.toml`, where geopandas and seaborn have lower bounds only) and exported to `requirements.txt`. The five notebooks were executed top to bottom from a clean kernel with `jupyter nbconvert`. The final clean run took 21.6 minutes in total, of which notebook 04 (tuning and bootstraps) took 14.5 minutes. Every random element uses seed 42, and the tables reproduce exactly across runs.

TABLE X
CLEAN RUN OF THE FIVE NOTEBOOKS (FIRST FIVE LINES OF `RESULTS/NOTEBOOK-RUN-TIMES.TXT`).

Notebook	Exit	Seconds	Finished
01-data-preparation	0	4	2026-10-10 20:13:57
02-exploratory-analysis	0	4	2026-10-10 20:14:01
03-clustering-risk-regions	0	10	2026-10-10 20:14:11
04-classification	0	871	2026-10-10 20:28:42
05-evaluation-and-interpretation	0	404	2026-10-10 20:35:26

Table X lists the clean run from `results/notebook-run-times.txt`; the later lines of that file are re-runs of notebooks 02, 03 and 05 for figure layout, and section 5 of notebook 05 (the sensitivity checks of Section V, computed from the saved predictions) was executed on its own. The output of `src.checks` follows. The executed notebooks keep their cell outputs, and `results/run-evidence.md` gives the environment listing. The frozen inputs (BIPAD snapshot of 19 September 2026, the frozen BIPAD geography, the ERA5 archive and the terrain and boundary files) are included with their `.meta.json` provenance so that steps 4 to 9 of the README run offline.

```
$ uv run python -m src.checks
panel (156849, 20), incidents (16790, 8), terrain (77, 5)
ok  panel reproduces the reference numbers
ok  feature lists: no loss_*/target columns,
    set A has no future_*, NOWX has no weather
ok  no loss_* columns in panel or incidents table
ok  set A unchanged when the future is scrambled
  259 recent-incident window counts recomputed by hand
ok  entry-lag rule: known-at-t counts, synthetic
    pre-2020 lags
ok  percentiles and target encoding use training
    seasons only
ok  leave-one-season-out target encoding for training rows
ok  persistence baseline uses known-at-t incidents only
  real ERA5 coverage of panel rows: 100.00%
  real ERA5 rain features on cached seasons:
    (171710, 14), mean 7-day rain 71.2 mm
ok  real-weather smoke test
all checks passed
```

Three long lines are wrapped at the indented continuation.

B. Features and Hyperparameters

Table XI defines every feature in sets A and B, and Table XII gives the hyperparameter grids, the settings chosen by leave-one-season-out tuning and their mean fold PR-AUC. Table XIII shows each chosen setting fold by fold. The spread across held-out seasons (SD about 0.03) is about twice the gap between the four main families, and fold skill broadly follows the season's base rate. For this reason the families are treated as close. The grids are small, and several chosen settings lie at a grid edge (for example $C = 0.3$ for the RBF SVM and $C = 0.01$ for logistic regression), so the comparison holds within the grids searched. Table XIV turns the 5% budget on the test seasons (1,625 alerts) into counts.

C. Leakage Audit

`src/checks.py` runs eight groups of assertions. Its output is listed above, and the written audit is `results/leakage-audit.md`. The checks are: (1) the panel reproduces the reference numbers (146,223 rows, 6.81% positive,

TABLE XI
FEATURE SETS. SET A IS THE FIRST FIVE GROUPS, SET B ADDS THE LAST GROUP, AND NOWX DROPS THE RAIN GROUP.

Group	Feature	Definition
Rain (A)	rain_1d	ERA5 precipitation on day t (mm)
	rain_3d	Sum over $t-2$ to t (mm)
	rain_7d	Sum over 7 days (mm)
	rain_14d	Sum over 14 days (mm)
	rain_30d	Sum over 30 days (mm)
	rain_api	Antecedent precipitation index, decay 0.9
	rain_hours_1d	Hours with precipitation on day t
	tmax_1d	Maximum 2 m temperature on day t (°C)
	rain_3d_pct	Percentile of the 3-day sum in the district's 2017 to 2022 distribution
Season	doy_sin	Day of year, sine
	doy_cos	Day of year, cosine
	month	Calendar month
Terrain	elev_mean	Mean ward elevation (m)
	elev_relief	Ward elevation range (m)
	share_above_1000m	Share of wards above 1,000 m
	share_above_3000m	Share of wards above 3,000 m
Recent incidents	recent_dam_3d	Damaging incidents known at t, dated $t-3$ to t
	recent_dam_7d	Same, $t-7$ to t
	nbr_recent_dam_3d	Same as recent_dam_3d, summed over neighbouring districts
	nbr_recent_dam_7d	Same, 7 days
	recent_flood_7d	Flood incidents, 7 days
	recent_landslide_7d	Landslide incidents, 7 days
	recent_heavy_rain_7d	Heavy-rainfall incidents, 7 days
	recent_lightning_7d	Lightning incidents, 7 days
History	te_dam	District × month rate of damaging days (shrunk, leave-one-season-out)
	te_flood	Same, flood
	te_landslide	Same, landslide
	te_heavy_rain	Same, heavy rainfall
	te_lightning	Same, lightning
Future rain (B)	future_rain_3d	Observed ERA5 rain, $t+1$ to $t+3$ (mm); set B only
	future_rain_max_1d	Maximum daily rain, $t+1$ to $t+3$; set B only
	future_rain_hours_3d	Rain hours, $t+1$ to $t+3$; set B only

TABLE XII
HYPERPARAMETER GRIDS, CHOSEN SETTINGS AND MEAN LEAVE-ONE-SEASON-OUT PR-AUC (2017 TO 2022, SET A).

Model	Grid and fixed settings	Chosen	Fold PR-AUC
LR	C ∈ {0.01, 0.1, 1.0}; L2, class weight balanced, standardised inputs	C = 0.01	0.136
RF	min_samples_leaf ∈ {20, 100}; max_features ∈ {sqrt, 0.5}; 300 trees, class weight balanced-subsample	min_samples_leaf = 20, max_features = sqrt	0.150
HGB	learning_rate ∈ {0.03, 0.1}; max_leaf_nodes ∈ {15, 31}; 300 iterations, min leaf 50, L2 1.0, class weight balanced	learning_rate = 0.03, max_leaf_nodes = 15	0.148
SVM-L	C ∈ {0.001, 0.01, 0.1}; LinearSVC, class weight balanced, standardised	C = 0.1	0.135
SVM-RBF	C ∈ {0.3, 1.0}; gamma ∈ {scale}; RBF, γ = scale, 12,000-row stratified subsample	C = 0.3, gamma = scale	0.118

split rates 6.65, 4.44 and 8.46%, 2026 check 10,626 rows at 5.78%); (2) no loss_⋆, target, same-day outcome or identifier column is a feature, set A has no future_⋆ column and NOWX has no weather column; (3) set A is unchanged when all weather after an arbitrary cutoff is scrambled and all incidents not yet entered at the cutoff are removed; (4) recent-incident counts never include an incident before its entry day, pre-2020 synthetic lags match the 2020 to 2025 distribution, and 259 window counts recomputed by hand agree; (5) rain percentiles and history rates do not change when test-season data are altered; (6) a training row's history encoding does not change when its own season's outcomes are altered, and test rows use all six training seasons; (7) the persistence baseline uses known-at-t incidents only; (8) with the full ERA5 archive no feature is missing.

TABLE XIII
LEAVE-ONE-SEASON-OUT PR-AUC OF EACH CHOSEN SETTING BY HELD-OUT SEASON (SET A), WITH THE SEASON'S POSITIVE RATE.

Held out	Base (%)	LR	RF	HGB	SVM-L	SVM-RBF
2017	5.9	0.144	0.151	0.160	0.139	0.124
2018	6.8	0.119	0.119	0.128	0.120	0.120
2019	6.5	0.169	0.187	0.174	0.167	0.120
2020	8.2	0.173	0.193	0.194	0.173	0.137
2021	7.1	0.128	0.139	0.140	0.126	0.106
2022	5.4	0.086	0.111	0.089	0.085	0.098
Mean		0.136	0.150	0.148	0.135	0.118
SD		0.033	0.034	0.037	0.033	0.014
Min		0.086	0.111	0.089	0.085	0.098
Max		0.173	0.193	0.194	0.173	0.137

TABLE XIV
ALERT OUTCOMES AT THE 5% BUDGET ON TEST 2024 TO 2025 (1,625 ALERTS, 2,749 POSITIVES, 32,494 DISTRICT-DAYS), DERIVED FROM POD AND THE ALERT RATE.

Model	Hits	False alarms	Misses	Correct neg.
Clim.	317	1,308	2,432	28,437
LR	390	1,235	2,359	28,510
RF	326	1,299	2,423	28,446
HGB	360	1,265	2,389	28,480
SVM-L	387	1,238	2,362	28,507
SVM-RBF	331	1,294	2,418	28,451
RF, set B	472	1,153	2,277	28,592

Three residual points are stated for completeness. Within the tuning folds, the rain percentiles and the training rows' history encoding were fitted once on all six training seasons, so a held-out tuning season is not fully unseen; this can flatter fold scores but cannot reach the test seasons. The synthetic entry lags are drawn from the entry-date metadata of 2020 to 2025 incidents, test seasons included; no label enters them. Year is not a feature of any model. The year-only control has no district term, so its position below climatology says nothing about leakage. The proper check adds year to a control on the shrunk district-by-month history rate and month: test PR-AUC moves from 0.136 to 0.138 (paired difference +0.002, +0.000 to +0.004), and the fitted year coefficient is close to zero.

D. Additional Figures and Tables

Figs. 7 to 13 and Table XV support the exploratory findings, the risk regions and the ablations reported in Section V.

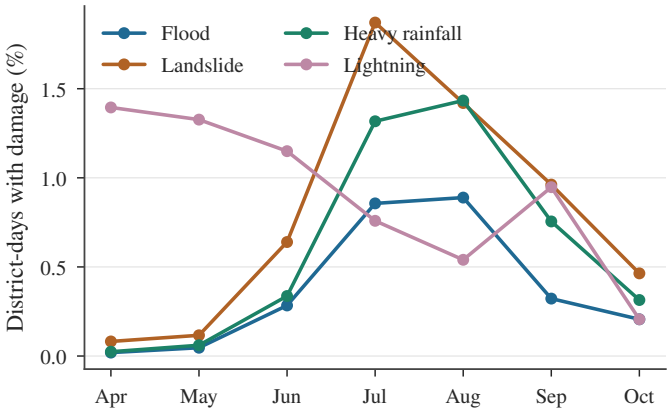


Fig. 7. Share of district-days with a damaging incident of each hazard by month, 2017 to 2025.

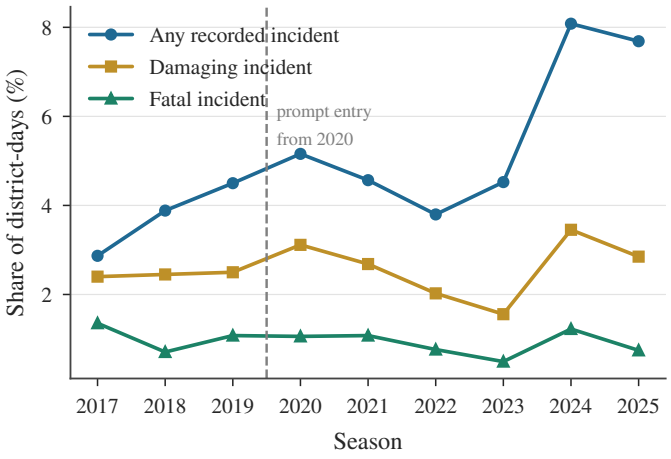


Fig. 8. Share of district-days per season with any recorded incident, a damaging incident and a fatal incident. Growth in "any" incidents reflects reporting as much as weather.

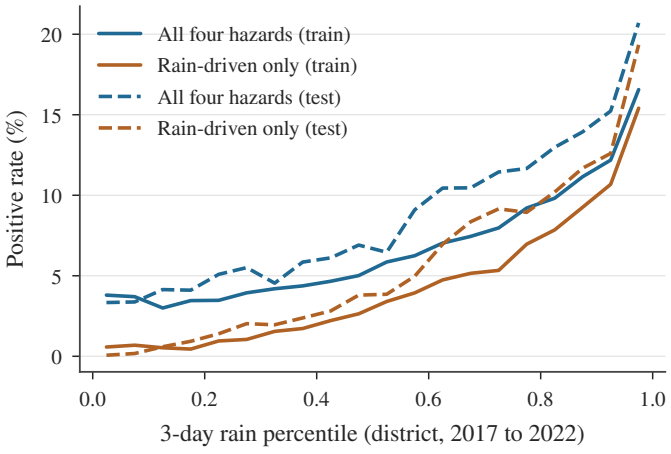


Fig. 9. Positive rate against the 3-day rain percentile (district, 2017 to 2022) for the all-hazard and rain-driven targets, in training and test seasons.

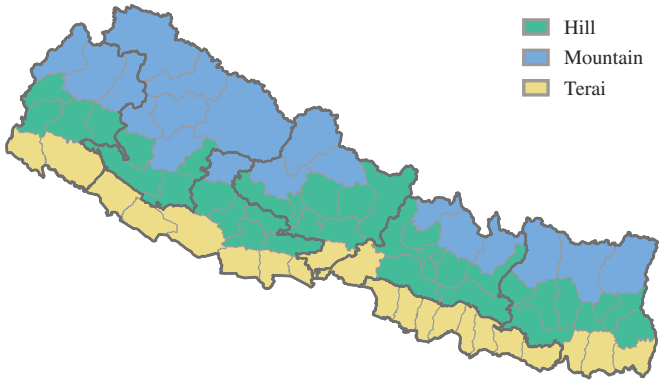


Fig. 10. Ecological belt proxy derived from ward elevations (Section III-A).

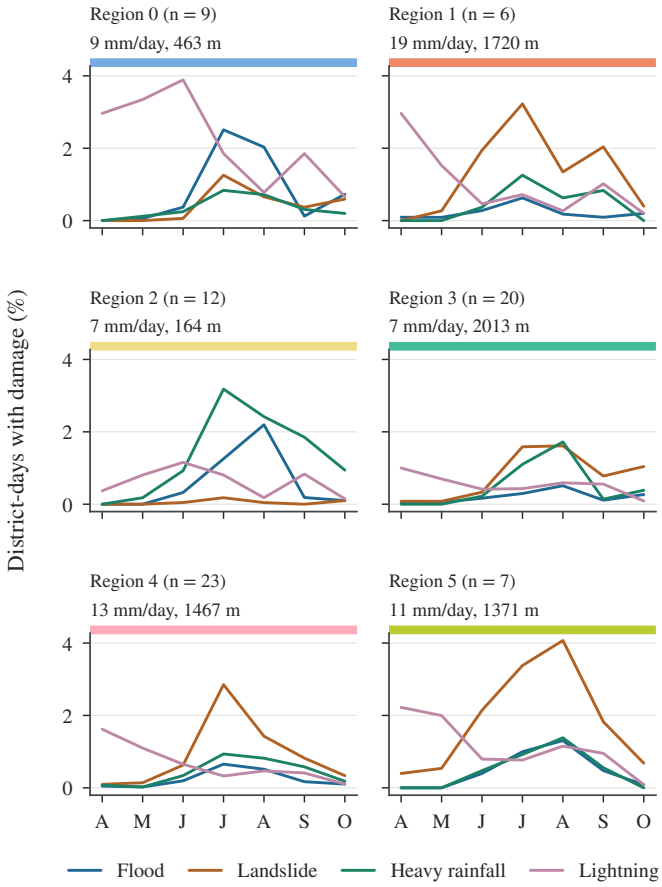


Fig. 11. Monthly rate of damaging days per hazard in each region, 2017 to 2022, with the region's size, mean daily rain and mean elevation.

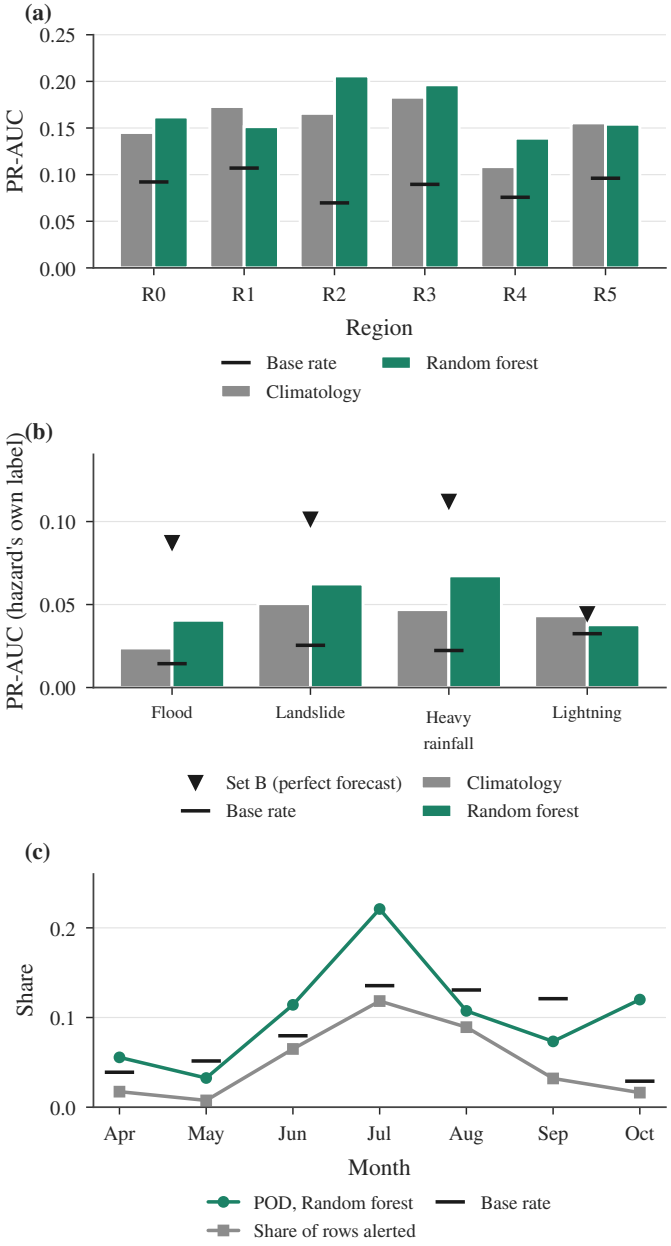


Fig. 12. Random forest (set A) on test 2024 to 2025. (a) PR-AUC by region against climatology, with base rates. (b) PR-AUC against each hazard's own label, with the set B ceiling. (c) Probability of detection and share of rows alerted by month at the national 5% budget.

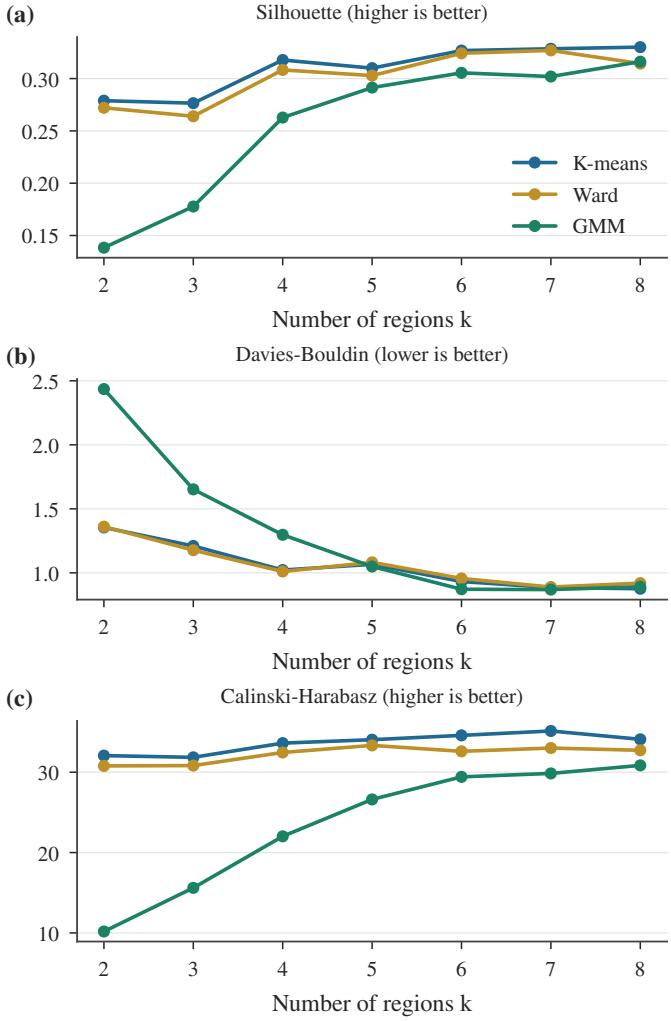


Fig. 13. Validity indices against k for the three clusterers in the four-component space.

TABLE XV
PR-AUC ON TEST 2024 TO 2025 BY FEATURE SET (NOWX: NO WEATHER; SET B ADDS THE OBSERVED RAIN FOR $t+1$ TO $t+3$), WITH PAIRED DIFFERENCES AND 95% WEEKLY BLOCK-BOOTSTRAP INTERVALS.

Model	NOWX	A	B	A − NOWX [95% CI]	B − A [95% CI]
LR	0.164	0.177	0.212	+0.014 [-0.005, +0.041]	+0.035 [+0.009, +0.072]
RF	0.143	0.165	0.219	+0.022 [+0.001, +0.045]	+0.054 [+0.014, +0.112]
HGB	0.164	0.175	0.220	+0.011 [-0.007, +0.030]	+0.045 [+0.010, +0.099]
SVM-L	0.164	0.177	0.211	+0.013 [-0.007, +0.040]	+0.034 [+0.009, +0.071]
SVM-RBF	0.141	0.153	0.179	+0.012 [-0.001, +0.030]	+0.026 [+0.009, +0.051]

ACKNOWLEDGMENT

Incident data are from the BIPAD portal of the National Disaster Risk Reduction and Management Authority, Government of Nepal. Weather and elevation data are from Open-Meteo (CC BY 4.0), based on ERA5 from the Copernicus Climate Change Service and the Copernicus DEM. Administrative boundaries are from OCHA through the Humanitarian Data Exchange (CC BY-IGO). Tools used are declared as the module leader instructs.

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